Learning-based spin estimation of table tennis ball with an event camera

Progress report week 14

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Summary

- SpinDOE dataset is now fully preprocessed and usable
- The dataset contains 2 recordings for every (143) ball gun setting. For every setting, the ground truth is measured using SpinDOE. Every trajectory was cut out of the event-recording and the ROI is extracted using hand-labeled positions.
- Labeled SpinDOE dataset into 6 labels (top-/backspin) (slow, mid, fast). Real data has rps up to 135rps and simulation only 80rps.
- tested model on real data. Very bad results see image.

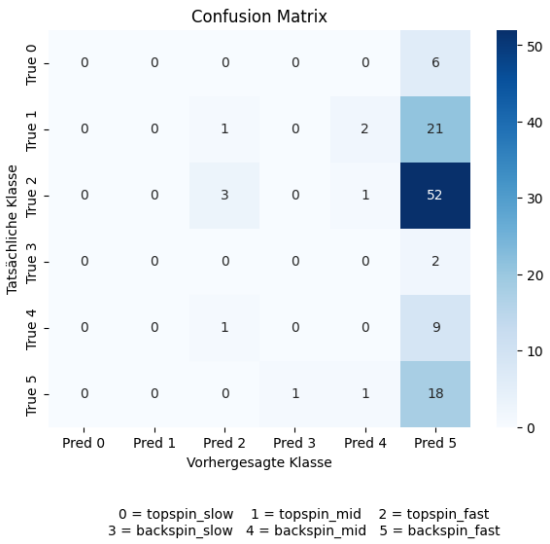

Figure 1: Confusion matrix

- Main problem: sim data has trahectory from right to left and real the other way around. I will try to flip and train a new model until tomorrow

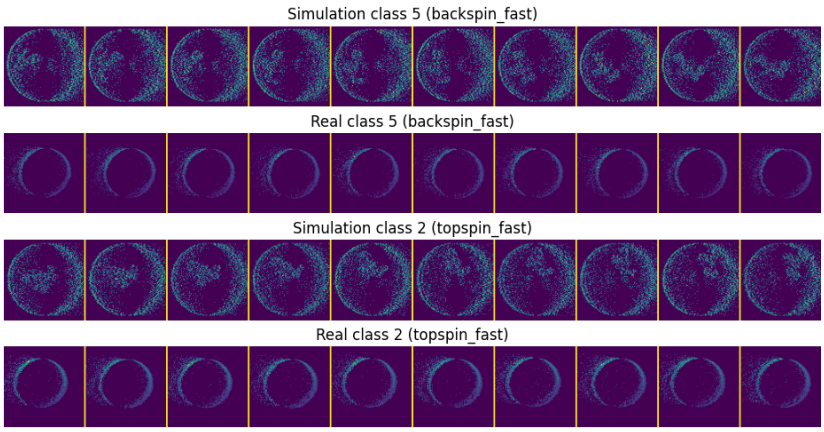

Figure 2: Real vs sim Data

Open questions

- How could the data be improved? trajectory instead of straight line? faster/slower ball? different scale?
- Look at examples of sim vs real together.

Next steps

- improve simualtion
- train new model